

1. Purpose and status

This document records the benchmark we intend to run later. It is a proposal and freeze candidate only. No benchmark instances are to be generated, trained on, executed, timed, or submitted to a prover as part of preparing this document.

The purpose of NOTALD is implementation validation. The benchmark deliberately presents the ALD with the negation of a theorem whose proof horizon is externally known. The system is therefore pointed at the wrong polarity on purpose. A correct implementation should recover by having the Refuter find the theorem on the opposite side, while the Prover is assigned a non-theorem side and must not manufacture a false proof. The benchmark is also intended to measure transfer from tied lower-resolution problems to larger related problems under a common learning protocol.

2. Notation correction

We reserve P for the Prover role and R for the Refuter role. To avoid using P for two different things, the Ocean problem family is written $\square(L)$, with a calligraphic P.

L is a number: the externally established minimum proof length, or proof horizon, under the frozen Ocean inference metric. $\square(L)$ is the problem instance having that horizon.

Therefore the corrected example is:

$L = 2.5 \times 10^3 = 2500$,
so $\square(L) = \square(2.5 \times 10^3) = \square(2500)$.

It is not correct to write $L = \square(2500)$, because L is a number and $\square(2500)$ is a benchmark problem.

3. Background: the transparent calibration theorem

$F(n)$

The transparent calibration family is defined by

$$\Gamma_n = \{p_0\} \cup \{\neg p_i \vee p_{i+1} : 0 \leq i < n\},$$

with target

$$F(n) = p_n.$$

Hypotheses are free, and one binary-resolution application counts as one inference. The direct unit-propagation proof derives $p_1, p_2, \dots, p_n$ in exactly n counted inferences.

The lower bound is also n . Every implication link $\neg p_i \vee p_{i+1}$ is necessary. If the link from p_k to p_{k+1} is removed, there is a valuation making p_0 through p_k true and all later atoms false while satisfying the remaining hypotheses and falsifying p_n . Thus every sound proof must depend on p_0 and all n implication clauses. A binary inference DAG with $n+1$ necessary leaves requires at least n binary inference nodes. Since an n -step proof exists and fewer than n steps are impossible,

$$L^*(F(n) \mid \Gamma_n) = n.$$

This is the clean calibration fact behind the Ocean benchmark. $F(n)$ is the transparent chain; it should not be confused with the hidden-route Ocean problem.

4. The hidden-route Ocean problem $\square(L)$

For the Ocean family, take a frozen directed graph $G = (V, E)$ with source s and target t . Encode the source as the fact p_s . Encode every directed edge $u \rightarrow v$ as the clause $\neg p_u \vee p_v$. The theorem target is $T_L := p_t$.

A native binary-resolution proof corresponds to following a directed path from s to t . Before the ATP sees the problem, the benchmark generator and an independent graph check establish that the shortest source-to-target path has length exactly L :

$$d(s, t) = L.$$

Under the frozen metric this gives

$$L^*(T_L \mid \Gamma_L) = L.$$

No ATP can return a valid native Ocean proof shorter than L , because that would encode a source-to-target route shorter than the independently verified shortest path. This is a property of the frozen problem, not a timeout claim and not an empirical guess about an ATP.

The Ocean can contain dead ends, detours, misleading branches, parallel corridors, and other search burden while preserving the same shortest proof horizon L . The trick ATP, Depths-F, is retained as a calibration control because it is specialized to this implication/reachability structure and largely removes the navigation problem. It is not to be interpreted as a fair general-purpose ATP competitor.

5. NOTALD: deliberate wrong-polarity evaluation

For each Ocean theorem T_L in $\square(L)$, NOTALD is not given T_L as its conjecture. Instead it is given

$C_L := \neg T_L$.

The benchmark designer knows that T_L is derivable with minimum native proof length L , but the running system is not given the expected status, the shortest route, the answer label, or an informative filename.

Using P for Prover and R for Refuter, P searches for $\Gamma_L \vdash C_L$, that is, $\Gamma_L \vdash \neg T_L$. R searches for $\Gamma_L \vdash \neg C_L$. After a frozen classical normalization of the double negation, R 's target is the native theorem $\Gamma_L \vdash T_L$.

Thus R is the role expected to detect theoremhood in NOTALD. P is deliberately assigned the non-theorem side. However, P failing or timing out is not itself a certificate of non-theoremhood. The benchmark is successful only when a status-appropriate verified certificate is produced.

For these frozen consistent Ocean instances, the expected final classification of C_L is REFUTABLE, or REFUTED if the current implementation uses that display label, because R should produce a verified proof of T_L . INDEPENDENT is wrong because T_L is provable. INCONSISTENT or CONTRADICTION is also wrong unless the base hypotheses themselves admit a verified contradiction. Budget exhaustion without a conclusive certificate must remain UNKNOWN or BOUNDED_UNKNOWN.

The point of NOTALD is not to trick the verifier. It is to test whether the implementation can recover from a deliberately wrong-polarity conjecture without being told which agent is supposed to win.

6. Massive tied benchmark grid

The proof horizons are

$$L = c \times 10^n,$$

where $n \in \{1, 2, 3, 4, 5\}$ and $c \in \{1, 2.5, 7.5\}$.

This gives 15 benchmark depths:

$n = 1$: 10, 25, 75
 $n = 2$: 100, 250, 750
 $n = 3$: 1,000, 2,500, 7,500
 $n = 4$: 10,000, 25,000, 75,000
 $n = 5$: 100,000, 250,000, 750,000

For example, $L = 2.5 \times 10^3 = 2500$, so the associated Ocean instance is $\square(2.5 \times 10^3) = \square(2500)$, and NOTALD receives the negated conjecture $C_{2500} = \neg T_{2500}$. The correct opposing certificate is a proof of T_{2500} whose native Ocean proof cannot contain fewer than 2500 counted edge-resolution inferences.

7. Tied-to-each-other learning protocol

The 15 problems form one deliberately related benchmark family. Their relationship is known to the benchmark designer but is not to be advertised to the systems through answer-bearing filenames, comments, L^* annotations, shortest-path metadata, or target-specific hints. The systems may infer regularities from permitted training data.

For a held-out evaluation problem $\square(L)$, every learning ATP or ALD must receive the same permitted lower-resolution tied training set. The natural tied set consists of benchmark siblings with smaller L . This is intended to measure whether learning on lower-resolution relatives improves head-to-head performance on larger relatives.

The exact training artifact bundle must be frozen before execution. In particular, before the run we must decide whether lower-resolution access includes only problem files, verified certificates, search traces, shared-bank events, or some specified subset. Whatever rule is chosen must be identical for every learning system to which it applies.

The held-out evaluation instance itself, its answer label, its shortest route, and its target-specific proof certificate must not be placed in training. The relation among the tied problems should be learnable from legitimate prior experience, not disclosed as an answer channel.

8. Systems in the head-to-head comparison

The proposed comparison includes the trick ATP / Depths-F calibration control, Data-ATP, Data 2.0.1, the four professional ATPs already selected for the benchmark suite, and the NOTALD system.

If a professional ATP is not a learning system, the tied-learning rule does not artificially add learning to it. It receives the same frozen evaluation problem and the resource protocol appropriate to its class. Exact software versions and the identities of the four professional ATPs must be frozen in the final run manifest.

9. Blinding, fairness, and implementation checks

Before execution, every evaluation instance should be sealed so that the system cannot read the intended answer from presentation details. The final protocol should remove L*, expected-status, and generator-answer comments; use opaque instance identifiers; neutralize informative symbol names where the existing Ocean protocol permits it; shuffle presentation order without changing the formal graph; keep the answer key and independently verified shortest path outside the solver process; and freeze verifier, parser, normalization, resource accounting, seeds, training exposure, and hardware rules before the first scored run.

The benchmark is designed to test logical polarity recovery, implementation safety, depth scaling, lower-resolution transfer, and shared-memory behavior. P must not generate a false proof of $\neg T_L$. R should eventually return the valid opposing proof of T_L when resources suffice. Independence and Inconsistency must not produce false settlements. If any agent produces incompatible certified evidence for the same frozen instance, that is an implementation or audit alarm, not a normal benchmark outcome.

10. Planned reporting and pre-run freeze items

The final scored benchmark should report, per system and per depth, the final certified status, correctness, charged search expansions to the first accepted conclusive certificate, returned proof or certificate length, L/L* for native Ocean proofs, per-agent work for NOTALD where available, wall-clock time, peak memory, shared-lemma production and cross-role reuse where applicable, BOUNDED_UNKNOWN separately from logical failure, and every invalid certificate or contradictory evidence as an audit failure.

Head-to-head rates should be computed only after the run protocol, tie rules, budgets, and handling of UNKNOWN are frozen.

Before any execution we still need to freeze the number of independently generated instances or seeds at each of the 15 depths; the exact four professional ATPs and software versions; global resource budgets and stopping rules; the lower-resolution training artifact bundle; whether all smaller-L siblings or only specified tied predecessors are exposed; the normalization rule for $\neg T_L \rightarrow T_L$ without adding artificial Ocean proof cost; the verifier and proof-counting convention; the blinded file-generation and answer-key procedure; and the scoring rule for head-to-head comparisons.

Until those items are frozen, this document is a benchmark proposal, not a run authorization.

Technical background

The transparent $F(n)$ definition, its exact n -inference proof and lower bound, and the hidden-route Ocean generalization come from the existing Proof Horizon and Ocean technical notes. Those notes distinguish the transparent $F(n)$ calibration chain from the hidden-route Ocean family and define the Ocean proof horizon from an independently verified shortest source-to-target path. The NOTALD polarity-reversal design in this proposal applies that calibrated family to the current multiagent ALD architecture.